Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**A technology that non-invasively detects battery internal defects utilizing an electromagnetic sensor array and a 2D imaging method with an image capture speed of less than 1sec**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Battery	*Manufacturing and Measurement/Evaluation Equipment*		
Concept of Proposed Technology	**[Core Concept]** A non-invasive diagnostic system that visualizes internal battery defects (e.g., dendrites, metal particles) by detecting magnetic field distortions using a high-speed electromagnetic sensor array and AI imaging algorithms. **[Main Functions]** Real-time 2.5D imaging of internal short circuit precursors (< 3 sec). Non-destructive testing compatible with various cell form factors. **[Potential Uses]** Manufacturing: Zero-defect screening in Gigafactories (Inline QC). Maintenance: Safety diagnosis for EVs and ESS without pack disassembly. Circular Economy: Rapid safety grading of used batteries for reuse (Second-life ESS) or recycling decisions.			

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Rationale for Joint R&D Project	**[Need for R&D]** Detection of internal battery defects, such as Li precipitation, metal particles, and dendrites, requires bulky, expensive equipment, such as CT and X-ray systems. X-rays do not provide sufficient resolution to detect tiny defects. Although CT has sufficient resolution, it is bulky, expensive, and time-consuming to detect small battery defects, which prevents efficient analysis of the large numbers of batteries required for many useful applications, such as failure pattern and source analysis during battery development or manufacturing yield stabilization. Consequently, there is a strong demand for an efficient, fast, yet accurate device for internal battery defect detection that can be used for battery development and testing, manufacturing quality control, and yield stabilization. **[Need for Joint Collaboration]** The proposed work aims to develop a non-intrusive, fast, and accurate battery internal defect detection device utilizing: • Magnetic battery sensing technology • Magnetic battery sensor array and sensor array signal processing technology • 2D magnetic image-based defect identification technology Purdue University, Collaboration institute of USA, has expertise in detecting battery defects (Li precipitation, metal particles, dendrites, etc.) utilizing magnetic field sensors and magnetic field map analysis. Led by CARES (Center for Advances in Resilient Energy Storage), a research hub jointly established by Purdue University and the Electrochemical Safety Research Institute (ESRI), Purdue conducts systematic fundamental and applied research on the safety, sustainability, and durability of next-generation energy storage systems. CARES covers a wide range of advanced energy storage technologies, including solid-state batteries, metal/liquid electrolyte systems, and sodium-ion batteries, and continuously enhances the industrial applicability of its research outcomes through the operation of application-oriented testbeds. Centered around CARES, Purdue University has established strong industry–academia–government collaboration networks and well-structured technology transfer systems, including the Purdue Research Foundation. Based on this foundation, Purdue has secured numerous core technologies and intellectual property rights in the battery field and maintains strong competitiveness in the commercialization and industrial dissemination of its research outcomes. The collaboration with a Korean SME with expertise in hardware and software integration, battery system integration, and battery testing will yield strong technological and commercial outcomes. Also, the collaboration with a Korean SME capable of working with major Korean battery manufacturers will be a significant advantage for the efficient validation of the developed technology and for accelerating its commercialization, both domestically and worldwide.
Key Words	**1. Battery defects (Li precipitation, metal particles, dendrite)** **1. Internal short circuit (ISC)** **3. Non-invasive battery defect detection** **4. Magnetic battery sensing** **5. Magnetic sensor array**

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	**Detailed Objectives** **1. High-Resolution Electromagnetic Sensor Array Module (HW)** - Development of a high-density magnetic sensor array optimized for EV battery cells to detect micro-magnetic field distortions. - Design of a noise-shielding structure to minimize environmental magnetic interference during the scanning process. **2. AI-Driven 2.5D Imaging & Defect Diagnosis Algorithm (SW)** - Development of a "Magnetic-to-Image" reconstruction algorithm that converts magnetic flux signals into 2.5D visual images in real-time. - Implementation of an AI/ML-based classification logic to distinguish specific defect modes (e.g., Li-dendrites, metal particles, internal short circuits) from normal battery variations. **3. High-Speed Non-Invasive Diagnostic System Integration (System)** - Development of a high-speed Data Acquisition (DAQ) system capable of processing multi-channel sensor signals within 1 second. - Integration of HW/SW into a product-level prototype (Tester) applicable to battery manufacturing lines and reuse inspection centers.			
R&D Target Specifications	**The target specifications** are as follows: • Sensing array size: up to 16 x 39 (624 sensors, pitch size >5 mm) • Maximum battery size for imaging = 17.5 cm x 41.5 cm • Imaging speed: < 3sec • Detectible defect size: at least 75 μm (diameter) • Gap (between battery and sensing array) tolerance: up to 3mm			
R&D Plan	1. Research Approach (Two-Track Strategy) To maximize efficiency and synergy, the project will be executed using a "Core Tech vs. Productization" two-track strategy: Track A (Purdue Univ.): Focuses on the "Core Physics & Algorithm," including magnetic field modeling, sensor array configuration, and 2.5D image reconstruction logic. Track B (Korean SME): Focuses on "System Integration & Validation," including High-Speed DAQ hardware design, commercial-grade housing, and field testing with Korean battery manufacturers. **2. Detailed Plan & Key Tasks (Year 1: Core Technology & Prototype)** Goal: Secure core sensing technology and verify basic performance (TRL 3 → 5). Key Tasks: [Purdue] Sensor Array Simulation & Design: Optimization of Hall-effect/MR sensor arrangement using FEA (Finite Element Analysis) to maximize sensitivity to internal shorts. [SME] High-Speed DAQ Board Development: Designing a multi-channel signal processing board			

capable of handling data acquisition within 1 second.
[Joint] Baseline Algorithm Development: Developing the initial logic to convert magnetic flux density (B) into 2D visual maps.
[SME] Jig & Fixture Fabrication: Manufacturing test jigs for various battery form factors (Pouch/Prismatic) to ensure stable scanning distance (Gap < 2mm).

3. Detailed Plan & Key Tasks (Year 2: System Integration & Validation)

Goal: System integration, SW optimization, and reliability testing (TRL 5 → 7).
Key Tasks:
[Purdue] 2.5D Imaging & AI Classification: Upgrading the algorithm to 2.5D (Depth estimation) and training AI models to classify defect types (Dendrite vs. Particle).
[SME] All-in-One Tester Integration: Integrating the sensor module, DAQ, and PC into a single industrial cabinet (Safety & Noise shielding applied).
[SME] GUI Development: Developing user-friendly software for production line operators (Pass/Fail indicator).
[Joint] Performance Validation: Conducting comparative tests against "Ground Truth" data (CT/X-ray) using defective cell samples provided by battery makers.

4. Methodologies

Magnetic Field Simulation: Utilization of electromagnetic simulation tools (e.g., Maxwell, COMSOL) to predict magnetic field distortions caused by micro-defects ($50\mu m$).
AI/ML Signal Processing: Implementation of Convolutional Neural Networks (CNN) for image pattern recognition to distinguish noise from actual defects.
Cross-Validation: Verification of defect detection accuracy by disassembling the battery (Destructive Analysis) after scanning to confirm the presence of dendrites.

Expected Milestones & Timeline

Timeline	Milestone (Key Deliverables)
Month 6	• Selection of optimal sensor type & array configuration (Design Freeze) • Completion of initial magnetic field simulation Report
Month 12	• Development of Alpha-Prototype (Sensor + Lab-scale DAQ) • Success in detecting artificial defects in dummy cells (PoC)
Month 18	• Completion of Beta-Prototype (Integrated System) • Implementation of Real-time (<1 sec) Signal Processing Logic
Month 24	• Final System Performance Test Report (Accuracy & Speed) • Field Test Result with Battery Manufacturer (Verification of TRL 7)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**[Roles and Responsibilities of Overseas Research Institution – Purdue University]** • Design an electromagnetic 2.5D imaging system hardware and build a prototype • Develop a 2.5D sensor array firmware • Develop a 2.5D electromagnetic imaging software • Develop a 2.5D image-based battery defect (Li precipitation, internal short circuit, etc.) diagnosis method • Development of high-resolution magnetic sensor array simulation & modeling.

	• Optimization of defect classification algorithms (AI/ML) for Li-precipitation and Dendrites. • Verification of the sensing limit through lab-scale experiments.
Expected R&R of Korean SME Partner	**[Roles and Responsibilities of Korean SMEs]** • Design and fabrication of the High-Speed DAQ (Data Acquisition) Hardware. • Development of the System GUI and integration of Purdue's algorithm. • Prototype manufacturing and field testing with Korean battery makers • Optimize the prototype design for production and quality control • Develop a scanning module for 2D to 2.5D conversion • Test 2.5D electromagnetic battery imaging and diagnostic system • Product level integration of the hardware, firmware, and software • Collaborate with battery manufacturers for validation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Existing non-invasive battery internal defect inspection equipment, such as CT and X-ray, is bulky, expensive, and time-intensive. The outcomes can lead to new, cost-effective yet accurate battery internal defect (Li precipitation, metal particles, dendrites, etc.) test equipment that can be widely used in labs, testing, and manufacturing facilities. - An equipment that can detect battery internal defects (Li precipitation, metal particles, dendrites, etc.) within 1 sec. - Cost-effective battery test equipment that can complement expensive equipment such as CT and X-ray. - Test equipment that can detect battery internal defects much faster than existing methods, such as CT and X-ray - A test equipment that can determine whether to reuse or recycle batteries The outcomes can result in new manufacturing equipment that can detect faulty batteries during manufacturing: - Improve manufacturing yield through efficient quality control equipment that can detect battery defects induced during battery manufacturing **Technological Outcomes: Overcoming Limits of Legacy Tech** • World-First Inline Inspection Solution: Securing proprietary technology for high-speed (< 1 sec) battery internal defect diagnosis that overcomes the speed limit of existing CT/X-ray (which takes > 30 min), enabling full-inspection (100%) on mass production lines. • Advanced Defect Visualization: Successfully developing a 2.5D magnetic imaging system capable of detecting specific failure modes such as Li-dendrites and micro-shorts, which are difficult to identify with conventional voltage/current monitoring (BMS). **Economic & Industrial Impact: Creating New Markets**

	• Yield Improvement in Gigafactories: Significantly reducing the "False Negative" rate in battery cell manufacturing. By detecting defects early, manufacturers can minimize waste and reduce costly recalls (e.g., GM Bolt case). • Revitalizing the Used Battery (UB) Market: Solving the bottleneck in the reuse/recycling industry by providing a rapid grading solution. This allows quicker decision-making on whether to Reuse (for ESS) or Recycle (for raw materials), accelerating the circular economy. • Import Substitution: Replacing expensive, imported foreign inspection equipment (CT/X-ray) with a cost-effective domestic solution, reducing CAPEX for battery manufacturers. **Social Impact: Enhancing Public Safety** • Preemptive Fire Prevention: Directly addressing the growing "EV Phobia" by detecting internal short circuit precursors (dendrites) before thermal runaway occurs, thereby protecting human lives and property. • Strengthening Energy Security: Supporting the stability of the national energy grid by ensuring the reliability of Second-life ESS (Energy Storage Systems) made from reused EV batteries.